

Counts or Conditional Percents?

AP Statistics · Unit 2 · Topic 2.1 · B: 2026-10-15 · E: 2026-10-19 · TEACHER KEY

CED TETHER

- **VAR-1.D** Identify questions to be answered about possible relationships in data. [Skill 1.A]
- **VAR-1.D.1** Apparent patterns and associations in data may be random or not.

Essential question: When group sizes differ, which comparison reveals whether two categorical variables are related?

DOK-3 skill: 4.B · **FRQ pattern:** counts-vs-conditional-percents-are-the-variables-related

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

First take → rules + (a)–(c) → turn in

DOK: 1 → 3

Minutes: 45

Teacher does

- Show the board at the bell and collect a one-sentence first take without confirming a claim.
- Read the rules box; work (a) and (b). Allow about 10 minutes to finish (a)–(c).
- Ask for evidence linked to a claim. Collect the sheet and hand-score part (c) with E/P/I.

Students do

- Choose the more useful line of reasoning and cite one table detail.
- Use the rules box to name the variables and compare the within-group percentages.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

Adult role

Circulate and ask students to connect their choice to evidence. Score part (c) using the three required reasoning elements; do not require dropped extension work.

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Supports Rosa using 70 percent versus 50 percent
- **reasoning-2** (required) — Explains that unequal group totals make 75 versus 35 misleading
- **reasoning-3** (required) — Limits the result to association, with no demonstrated cause

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Dividing 35 and 75 by the grand total instead of their row totals
- Saying 75 is a larger percentage because it is a larger count
- Claiming that association proves playing a sport increases sleep

[STAR] TODAY'S PROBLEM — annotated

Suppose a counselor records whether each of 200 students plays a school sport and whether the student usually gets at least 8 hours of sleep on a school night.

Kai: *Seventy-five nonathletes but only 35 athletes get at least 8 hours, so playing a sport goes with less sleep.*

Rosa: *Those counts compare groups of different sizes; 35 out of 50 may tell a different story from 75 out of 150.*

Usual sleep on a school night

Sport?	≥ 8 h	< 8 h	Total
Plays	35	15	50
Does not	75	75	150
Total	110	90	200

FIRST TAKE (5 min)

Which student's reasoning seems more useful for deciding whether sport participation and sleep are related? Name one table detail.

Key: Not graded. Expect the 75-to-35 count comparison to attract students before they normalize for group size.

Rules callout "COMPARE WITHIN GROUPS (keep on the page)" is printed on the student sheet.

DOK 1 (a) Name the two categorical variables.

Key: The variables are school-sport participation and usual school-night sleep category.

DOK 2 (b) Find the percent who get at least 8 hours of sleep in each sport group. Show the two fractions.

Key: Athletes: $35/50 = 70\%$. Nonathletes: $75/150 = 50\%$.

DOK 3 (c) Whose reasoning is better supported? Use both percentages and explain why comparing 75 with 35 can mislead. State whether this table shows that playing a sport causes more sleep. Write 3–4 sentences.

Key: Rosa's reasoning is supported: 70% of athletes versus 50% of nonathletes get at least 8 hours. The count 75 comes from a group of 150, three times the athlete group of 50, so the larger count does not mean a larger within-group percentage. The table shows an association for these students, but observing their choices does not show that playing a sport caused more sleep.

EXIT

One thing the rules box changed about my first take: _____